

An Empirical Study of Low Precision Quantization for TinyML

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ABSTRACT

Tiny machine learning (tinyML) has emerged during the past few years aiming to deploy machine learning models to embedded AI processors with highly constrained memory and computation capacity. Low precision quantization is an important model compression technique that can greatly reduce both memory consumption and computation cost of model inference. In this study, we focus on post-training quantization (PTQ) algorithms that quantize a model to low-bit (less than 8-bit) precision with only a small set of calibration data and benchmark them on different tinyML use cases. To achieve a fair comparison, we build a simulated quantization framework to investigate recent PTQ algorithms. Furthermore, we break down those algorithms into essential components and re-assembled a generic PTQ pipeline. With ablation study on different alternatives of components in the pipeline, we reveal key design choices when performing low precision quantization. We hope this work could provide useful data points and shed lights on the future research of low precision quantization.

KEYWORDS

tinyML, neural networks, quantization, benchamrk

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1 INTRODUCTION

Embedding AI directly on edge devices is becoming a key to revolutionize the internet of things (IoT), where billions of tiny devices are leveraged to gain great productivity and efficiency in areas including consumer, medical, automotive and industrial. As a result, tiny machine learning (tinyML) has emerged over the past few years

that aims to deploy machine learning algorithms, especially neural networks, on embedded AI processors such as micro-controllers or embedded NPUs (e.g. Qualcomm's eNPU [23] and ARM's MicroNPU [1]) with very low power consumption in the level of a few milli-watts. By performing inferences on the devices, tinyML brings the advantage of high energy efficiency, fast responsiveness, high privacy and strong autonomy of edge devices. However, the embedded AI processors are usually highly resource-constrained with limited memory and computation capability. A large body of literature has focused on addressing these issues by making neural networks more efficient, including efficient neural architecture search [4, 18, 29], neural networks and system co-design [16, 17] and model compression such as pruning [10], knowledge distillation [11] and quantization [10, 14].

Neural network quantization is one of the most important model compression techniques. It compresses models by using low-bit precision representation for weight and activation tensors, instead of the 32-bit (or 16-bit for half-precision) precision that is commonly used during model training. The memory consumption and computation of a low precision model are greatly reduced. And quantization compresses a model without changing its architecture. This is particularly useful when the model architecture is already optimally designed for a specific inference system or device. Besides, quantization can be applied along with other model compression techniques, such as pruning and knowledge distillation.

Neural networks have been shown to be quite robust to quantization with 8-bit precision without sacrificing the accuracy[20]. Quantization with lower precision can further reduce the memory consumption and computation. And ultra-low precision (1 or 2-bit) operations can often be computed efficiently with bit-wise arithmetic and thus achieving signification computation acceleration [28]. However, due to the large quantization noise, the benefits of low precision quantization often come at the cost of significant accuracy degradation. Among approaches to mitigate the accuracy drop caused by quantization, post-training quantization (PTQ) directly performs quantization on a pre-trained full precision model to convert it to a corresponding low precision model. Unlike quantization-aware training (QAT), PTQ does not require the full training pipeline of the original model and can be applied with a small amount of calibration data (or even data-free [21]), thus becoming a popular model quantization routine.

*Equal contribution

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